

Beyond Aggregate Error Metrics: Revealing Hidden Error Structures in Graph-Based Traffic Forecasting

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Abstract. Accurate evaluation is fundamental to progress in spatio-temporal forecasting. Yet, traffic prediction models, such as DCRNN and AGCRN, are still predominantly compared using global error metrics such as MAE and RMSE, implicitly assuming that models with similar aggregate accuracy exhibit similar predictive behaviour. In this paper, we challenge this assumption. We investigate whether global metrics are sufficient to characterize the quality of graph-based traffic forecasting models. Rather than focusing on architectural improvements, we examine the hidden structure of prediction errors across space, time, and network topology. Our analysis combines multiple complementary perspectives, including residual temporal dependencies, graph-aware spatial error organization, spectral characteristics of residual signals, and the preservation of spatio-temporal traffic dynamics through a Graph-Temporal Operator Error (GTOE). Experiments conducted on several benchmark traffic datasets using widely used and representative graph-based forecasting models reveal a consistent and underexplored phenomenon: models that appear nearly equivalent according to conventional metrics can exhibit substantially different residual structures, dynamical behaviour, and graph-level error patterns. More importantly, these differences may alter the relative assessment of competing models, raising questions about the reliability of conclusions drawn from global metrics alone. Our findings suggest that forecasting accuracy and dynamical fidelity are not necessarily aligned objectives. Beyond traffic prediction, this study highlights a broader methodological issue in spatio-temporal machine learning: improvements measured by aggregate metrics may conceal important differences in how models represent and propagate complex system dynamics. We argue that understanding these hidden dimensions is essential for more reliable and trustworthy evaluation of graph-based forecasting systems.

Keywords: Traffic Forecasting · Graph-Based Learning · Spatio-Temporal Forecasting · Residual Analysis · Dynamical Consistency · Model Evaluation

1 Introduction

Traffic forecasting is a central task in intelligent transportation systems, supporting congestion management, route planning, and urban mobility optimization [14,25]. It also represents one of the most challenging applications of spatio-temporal machine learning. Traffic data are highly dynamic, graph-structured, noisy, and affected by recurrent and non-recurrent patterns such as peak hours, congestion propagation, local disruptions, and sensor imperfections [16,22,21,12]. These characteristics make traffic forecasting a demanding test case for evaluating whether learning models truly capture spatio-temporal behavior rather than only minimizing average prediction errors.

Graph-based spatio-temporal models have become a dominant paradigm for this problem. By representing traffic sensors or road segments as graph nodes, these models explicitly encode spatial dependencies induced by the transportation network. Representative architectures such as DCRNN, STGCN, Graph WaveNet, ASTGCN, and AGCRN have introduced diffusion processes, graph convolutions, attention mechanisms, adaptive graph learning, and node-specific representations to better capture complex traffic dependencies [3,14,1]. Most progress in the field, however, has focused on improving forecasting architectures.

In contrast, the evaluation of traffic forecasting models remains largely dominated by aggregate error metrics, especially mean absolute error (MAE), root mean square error (RMSE), and mean absolute percentage error (MAPE) [11]. These metrics are useful because they provide compact and comparable summaries of predictive accuracy. However, they collapse prediction errors across graph nodes, time steps, and forecasting horizons into global scores. As a result, they provide limited information about how errors are organized over the traffic network, whether they persist over time, or whether predictions preserve the spatio-temporal dynamics of the underlying system [8].

This limitation is particularly important in graph-based traffic forecasting. Prediction errors are not independent scalar deviations: they occur on a network, evolve over time, and may reflect different types of model failure [3]. Two models may obtain similar MAE or RMSE values while exhibiting different residual temporal dependencies, spatial error concentrations, graph-frequency behaviour, or dynamical inconsistencies [3]. In such cases, aggregate accuracy alone may provide an incomplete basis for assessing model reliability. Several alternatives to aggregate metrics have been explored in structured prediction contexts, including alignment-based measures for sequences and time series [18,5], geometric distances for trajectories and spatial sets [9,6], and optimal transport formulations for comparing distributions and structured objects [17]. Spatio-temporal residual dependence has also been studied using Moran’s I-based statistics [24,19], while dedicated residual diagnostics have been developed for space-time point processes [4]. However, these approaches primarily assess residual dependence or model fit, rather than complementary error structures across nodes, time, and graph topology.

In this paper, we revisit the evaluation of graph-based traffic forecasting models from a residual-structure perspective. Rather than introducing a new forecasting architecture, we analyze how prediction errors are organized across time, graph topology, and traffic dynamics. To this end, we combine temporal, spatial, spectral, and operator-based dynamical diagnostics to investigate whether models with similar aggregate accuracy exhibit similar forecasting behaviour. Our objective is to identify aspects of model performance that remain hidden when evaluation relies solely on global metrics such as MAE and RMSE.

The main contributions of this work are as follows:

- We investigate the limitations of aggregate evaluation metrics by analyzing residual structures in graph-based traffic forecasting models.
- We propose a multi-perspective diagnostic framework that integrates complementary temporal, spatial, spectral, and dynamical views, each revealing a distinct aspect of forecasting behaviour concealed by aggregate error metrics.
- We introduce the Graph-Temporal Operator Error (GTOE), an operator-based diagnostic that jointly assesses the preservation of temporal variations and graph-level spatial dynamics in predicted traffic fields.
- We conduct an extensive empirical study on benchmark PeMS datasets and show that models with similar MAE and RMSE can exhibit substantially different residual behaviour and ranking outcomes under complementary evaluation criteria.

These diagnostics are used as complementary views, each targeting a different limitation of aggregate metrics.

2 Residual Analysis Methodology

This section presents the residual analysis methodology used to examine whether aggregate error metrics are sufficient to characterize the behaviour of graph-based traffic forecasting models. We analyze the structure of prediction errors across temporal, spatial, spectral, and dynamical dimensions.

The diagnostics are deliberately selected as complementary views of residual behaviour, capturing temporal, spatial, spectral, and dynamical information concealed by aggregate metrics. Together, they allow us to determine whether models with similar MAE and RMSE exhibit similar forecasting behaviour. Accordingly, the methodology is organized into four complementary views: aggregate evaluation (Section 2.1), temporal residual dependence (Section 2.2), spatial and spectral residual organization (Section 2.3), and spatio-temporal dynamical consistency (Section 2.4).

2.1 Problem Setting and Aggregate Evaluation Metrics

We consider a traffic forecasting task defined over a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ denotes the set of traffic sensors and $\mathcal{E}$ represents spatial connections. Let $N =$

$|\mathcal{V}|$, T the number of time steps, and H the forecasting horizon. We define the residual as the pointwise difference between the observed and predicted values

$$R_{t,h,i} = Y_{t,h,i} - \hat{Y}_{t,h,i},$$

forming a spatio-temporal residual field $R \in \mathbb{R}^{T \times H \times N}$ that remains indexed by time, horizon, and graph node.

Standard aggregate metrics summarize this field into scalar scores used as reference criteria throughout this work:

$$\text{MAE} = \frac{1}{THN} \sum_{t=1}^T \sum_{h=1}^H \sum_{i=1}^N |R_{t,h,i}|. \quad (1)$$

$$\text{RMSE} = \sqrt{\frac{1}{THN} \sum_{t=1}^T \sum_{h=1}^H \sum_{i=1}^N R_{t,h,i}^2}. \quad (2)$$

$$\text{MAPE} = \frac{100}{THN} \sum_{t=1}^T \sum_{h=1}^H \sum_{i=1}^N \left| \frac{R_{t,h,i}}{Y_{t,h,i} + \epsilon} \right|. \quad (3)$$

While compact and comparable, these metrics apply uniform aggregation across all nodes, horizons, and time steps. Consequently, different residual configurations—uniformly distributed errors versus errors concentrated on specific sensors or persistent over time—may yield identical aggregate scores. The following sections analyze the residual field from complementary perspectives to examine what this aggregation conceals. Throughout this study, we focus on MAE and RMSE as the primary reference metrics. While MAPE is also commonly reported, it is known to be unstable when ground-truth values are close to zero [13]. More importantly, MAE and RMSE remain widely used for model comparison in traffic forecasting studies [3,11], making them the natural reference point for examining what standard evaluation practice reveals and conceals.

2.2 Temporal Residual Dependence

Aggregate metrics summarize the magnitude of forecasting errors, but they do not indicate whether these errors behave as independent residuals or preserve temporal structure. In time series forecasting, a well-specified model is expected to reduce systematic temporal patterns in the residuals. If residuals remain strongly autocorrelated, this suggests that part of the temporal dynamics of the underlying process has not been fully captured by the model.

For each model, we analyze the temporal dependence of the residual sequence using autocorrelation at different time lags. Let $R_{t,h,i}$ denote the residual at time step t , forecasting horizon h , and node i . For a given lag ℓ , the residual autocorrelation [2] is computed as

$$\rho(\ell) = \frac{\sum_{t=\ell+1}^T (\bar{R}_t - \mu_R) (\bar{R}_{t-\ell} - \mu_R)}{\sum_{t=1}^T (\bar{R}_t - \mu_R)^2}. \quad (4)$$

where $\bar{R}_t$ denotes the residual aggregated over nodes and horizons at time t , and μ_R is the mean of the residual sequence. This analysis follows the standard use of residual autocorrelation in time-series diagnostics, and is also motivated by recent traffic forecasting studies showing that residuals of deep spatio-temporal models may remain autocorrelated and structured [26].

This analysis answers whether forecasting errors are temporally isolated or persistent. Low autocorrelation indicates that residuals behave closer to random errors, while high autocorrelation suggests that errors persist over successive time steps. In traffic forecasting, such persistence may indicate that the model has not fully captured recurrent temporal patterns, congestion evolution, or delayed propagation effects. Therefore, residual autocorrelation provides information that is not directly visible from global MAE or RMSE values.

2.3 Spatial and Spectral Residual Organization

Aggregate error metrics do not indicate how residuals are distributed across the traffic network. We therefore analyze the residual field at two complementary levels: node-level error concentration and graph-level residual organization.

First, for each node i , we compute a local error score by averaging residual magnitudes over all time steps and forecasting horizons:

$$\text{MAE}_i = \frac{1}{TH} \sum_{t=1}^T \sum_{h=1}^H |R_{t,h,i}|. \quad (5)$$

The deviation of each node from the global error is then defined as

$$\Delta_i = \text{MAE}_i - \text{MAE}. \quad (6)$$

A positive value of Δ_i indicates that node i exhibits larger errors than the global average, while a negative value indicates lower-than-average errors. This allows us to identify whether residual errors are uniformly distributed over the network or concentrated around specific sensors.

Second, we examine graph-level residual organization. Following graph signal processing, where the Laplacian quadratic form is commonly used to measure signal smoothness over a graph [20,15], we treat $\mathbf{r}_{t,h} \in \mathbb{R}^N$ as a residual graph signal and compute:

$$S_{t,h} = \mathbf{r}_{t,h}^\top L \mathbf{r}_{t,h}. \quad (7)$$

Higher $S_{t,h}$ values indicate sharper residual variations across connected sensors, which correspond to high-frequency graph components in graph signal processing [20,15]. The High-Frequency Energy Ratio (HFER) therefore measures the proportion of residual energy in this high-frequency range, and its 90th percentile provides a robust upper-tail summary that captures recurrent residual bursts without relying on the maximum.

The spatial and spectral analyses complement aggregate accuracy metrics by indicating where residual errors concentrate and whether their graph-level organization is smooth or locally irregular.

2.4 Spatio-Temporal Dynamical Consistency

The previous analyses examine residual magnitude, temporal persistence, and spatial organization. However, a forecasting model may also be evaluated according to whether it preserves the evolution of the traffic signal itself. In other words, two models may obtain similar aggregate errors while differing in their ability to reproduce temporal changes and graph-level spatial variations. To examine this aspect, we compute an operator-based diagnostic quantity that compares the dynamics of the observed and predicted traffic fields.

This diagnostic evaluates whether the model preserves spatio-temporal dynamics by comparing observed and predicted signals after two transformations: temporal differencing for short-term changes and graph Laplacian filtering for local variations across connected sensors [20,15]. The objective is not to introduce a replacement for MAE or RMSE, but to complement them by assessing whether the prediction preserves basic temporal and graph-spatial operators applied to the traffic signal.

We first consider a temporal difference operator. For each node i and horizon h , the observed temporal variation is defined as

$$\Delta_t Y_{t,h,i} = Y_{t+1,h,i} - Y_{t,h,i}. \quad (8)$$

and the corresponding predicted temporal variation is

$$\Delta_t \hat{Y}_{t,h,i} = \hat{Y}_{t+1,h,i} - \hat{Y}_{t,h,i}. \quad (9)$$

This component evaluates whether the model preserves the direction and magnitude of temporal changes.

We then consider a graph-spatial operator based on the graph Laplacian L . For each time step t and horizon h , let $\mathbf{y}_{t,h} \in \mathbb{R}^N$ and $\hat{\mathbf{y}}_{t,h} \in \mathbb{R}^N$ denote the observed and predicted traffic signals over all nodes. The Laplacian-transformed signals

$$L\mathbf{y}_{t,h} \quad \text{and} \quad L\hat{\mathbf{y}}_{t,h}. \quad (10)$$

represent local graph-spatial variations induced by the network topology. Comparing these quantities allows us to assess whether the prediction preserves local contrasts between neighboring nodes.

As a diagnostic quantity, we compute the Graph-Temporal Operator Error (GTOE) as

$$\text{GTOE} = \alpha \frac{1}{(T-1)HN} \sum_{t=1}^{T-1} \sum_{h=1}^H \sum_{i=1}^N \left| \Delta_t Y_{t,h,i} - \Delta_t \hat{Y}_{t,h,i} \right| + \beta \frac{1}{TH} \sum_{t=1}^T \sum_{h=1}^H \frac{\|L\mathbf{y}_{t,h} - L\hat{\mathbf{y}}_{t,h}\|_1}{N}. \quad (11)$$

Here, α and β are non-negative coefficients controlling the relative contribution of temporal and graph-spatial consistency. Unless otherwise stated, we use $\alpha = \beta = 1$ in order to avoid additional tuning and to treat both components as complementary diagnostic terms.

The first term measures temporal consistency by penalizing differences between observed and predicted temporal variations. The second term measures

graph-spatial consistency by penalizing differences between observed and predicted Laplacian-transformed signals. Lower GTOE values indicate that the model better preserves the spatio-temporal evolution of the traffic field.

This analysis complements aggregate accuracy metrics. MAE and RMSE quantify the average magnitude of prediction errors, whereas GTOE evaluates whether the predicted signal behaves similarly to the observed signal under simple temporal and graph-spatial operators. Therefore, if two models obtain similar global accuracy but different GTOE values, this suggests that aggregate metrics alone do not fully characterize their ability to preserve traffic dynamics.

3 Experimental Setup

3.1 Datasets

Experiments are conducted on three benchmark datasets from the PeMS collection: PeMSD04, PeMSD08, and PeMSD07(M), summarized in Table 1. Nodes correspond to traffic sensors and edges encode spatial relationships between sensors.

Table 1: Summary of the traffic forecasting datasets.

Dataset	Target	Nodes	Time steps	Input	Horizon
PeMSD04	Traffic flow	307	16,992	12	12
PeMSD08	Traffic flow	170	17,856	12	12
PeMSD07(M)	Traffic speed	228	28,224	12	12

3.2 Forecasting Models

We evaluate six forecasting models: five representative graph-based architectures and one non-graph recurrent baseline. The graph-based models are Diffusion Convolutional Recurrent Neural Network (DCRNN) [14], Spatio-Temporal Graph Convolutional Network (STGCN) [25], Graph WaveNet [23], Adaptive Graph Convolutional Recurrent Network (AGCRN) [1], and Attention-Based Spatial-Temporal Graph Convolutional Network (ASTGCN) [7]; the Gated Recurrent Unit (GRU) is included as a non-graph temporal baseline. The objective is not to compare the newest architectures, but to analyze residual structures in established benchmark models.

3.3 Implementation and Evaluation

All experiments are conducted using LibCity [10], an open-source library for urban spatio-temporal prediction that provides standardized implementations, preprocessing, and train/validation/test splits across all models and datasets. Each model uses an input window of 12 time steps, a forecasting horizon of 12 steps, and is trained for 70 epochs with a batch size of 8. The default LibCity random seed (42) is used across all experiments, ensuring reproducibility and

comparability with prior work conducted under the same framework. For each trained model, predicted values $\hat{Y}$ and ground-truth values Y are stored on the test set and used to compute both aggregate metrics and the residual-based analyses described in Section 2.

4 Results and Discussion

This section reports the experimental results and analyzes what aggregate forecasting metrics reveal and what they may conceal. We first examine conventional MAE and RMSE scores at global and horizon-wise levels, then analyze residual behaviour through temporal, spatial, spectral, and dynamical perspectives.

4.1 Aggregate and Horizon-Wise Forecasting Performance

Table 2 summarizes both global and horizon-wise results for the three PeMS datasets. The columns S, M, and L correspond to short, mid, and long horizons, i.e., horizons 1–4, 5–8, and 9–12, while G denotes the global score over all forecasting horizons.

Table 2: Aggregate and horizon-wise forecasting performance on the PeMS datasets. S/M/L/G denote short, mid, long, and global horizons. Best values within each dataset are highlighted in bold.

Dataset	Model	S-MAE	M-MAE	L-MAE	G-MAE	S-RMSE	M-RMSE	L-RMSE	G-RMSE
PeMS08	AGCRN	14.755	16.463	17.809	16.342	22.777	25.640	27.705	25.374
	DCRNN	14.412	16.697	18.501	16.537	22.323	25.956	28.543	25.607
	GWNET	14.747	17.270	19.326	17.114	23.011	27.064	30.024	26.700
	STGCN	15.769	20.557	25.335	20.554	23.740	30.768	37.539	30.682
	ASTGCN	18.831	21.341	22.795	20.989	27.798	31.906	33.693	31.132
	GRU	21.319	22.095	23.024	22.146	33.818	34.971	36.476	35.088
PeMS04	GWNET	17.778	19.288	20.422	19.163	29.811	32.277	33.768	31.952
	AGCRN	18.392	19.711	20.511	19.538	30.235	32.754	34.162	32.384
	DCRNN	18.553	21.315	23.505	21.124	29.643	33.693	36.932	33.423
	ASTGCN	20.668	21.575	22.627	21.623	32.553	34.085	35.829	34.156
	STGCN	18.616	22.391	25.625	22.211	29.723	34.970	39.393	34.695
	GRU	22.548	22.673	22.806	22.676	37.948	38.085	38.241	38.091
PeMS07	GWNET	1.916	2.857	3.277	2.683	3.569	5.639	6.466	5.225
	AGCRN	1.967	2.837	3.219	2.674	3.641	5.685	6.508	5.278
	STGCN	1.980	3.114	3.787	2.960	3.581	5.977	7.225	5.595
	DCRNN	1.967	3.119	3.836	2.974	3.712	6.227	7.654	5.865
	ASTGCN	2.857	3.725	4.177	3.586	4.849	6.306	7.004	6.053
	GRU	3.715	3.766	3.791	3.758	6.967	7.143	7.227	7.112

The aggregate results provide the conventional benchmark view of model performance. Overall, graph-based models outperform the recurrent GRU baseline, confirming the importance of spatial dependencies in traffic forecasting. However, the best model is dataset-dependent: AGCRN achieves the lowest global errors on PeMS08 and PeMS07(M), whereas GWNET performs best on PeMS04.

The horizon-wise results show that errors generally increase with the forecasting horizon, but not at the same rate for all models. Short-horizon rankings may differ from global rankings, as observed with DCRNN on PeMS08 and PeMS04. These differences motivate the residual-based analyses that follow.

4.2 Temporal Residual Dependence

We next examine whether prediction errors behave as isolated deviations or preserve temporal structure. Figure 1 reports the residual autocorrelation curves for the evaluated models. As shown in Figure 1, residuals remain strongly autocorrelated at short lags across the datasets, indicating that forecasting errors are not purely random. The rate of decay, however, differs substantially across models. For instance, AGCRN and GWNET reduce residual dependence more rapidly on PeMS07(M), whereas ASTGCN and GRU exhibit more persistent residual correlations in several cases. This shows that models with comparable aggregate accuracy may still leave different amounts of temporal structure in their residuals.

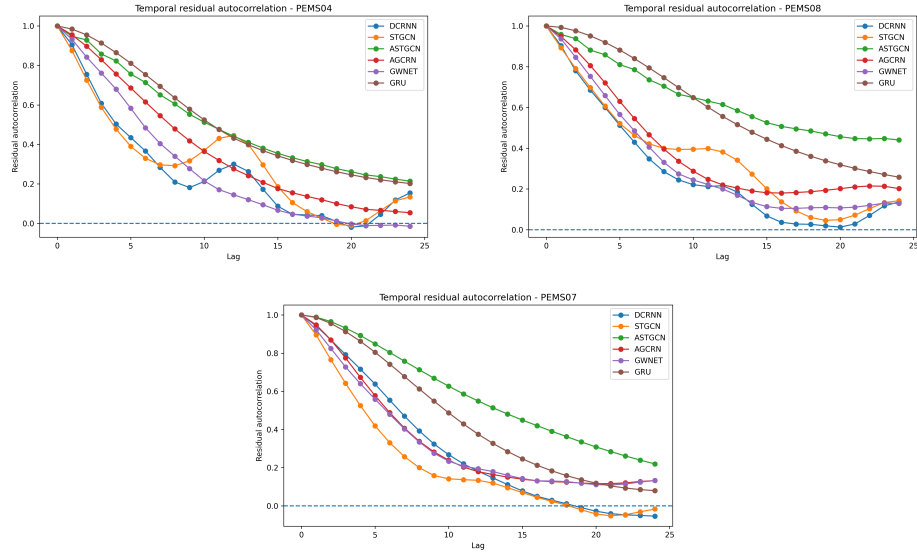


Fig.1: Temporal residual autocorrelation across PeMS04, PeMS08, and PeMS07(M).

In addition to autocorrelation, we examine whether forecasting errors increase under dynamically critical temporal conditions. We consider two such conditions: high traffic intensity, defined as observations above the 90th percentile, and large temporal variation, defined as changes in the observed signal above the 90th percentile. These contexts are important because traffic peaks and abrupt changes correspond to situations where forecasting reliability is particularly critical. Table 3 summarizes the resulting error amplification ratios. For each critical context, we compute a context-specific MAE using only the corresponding observations and report its ratio to the global MAE. A ratio above one means that errors are larger than average in that context, whereas a ratio below one means that the context is not more difficult than average. MAE is used because the objective is to measure average error amplification within each context, without giving additional weight to extreme errors as RMSE would do.

Table 3: Summary of temporal critical error amplification. Ratios correspond to context MAE divided by global MAE.

Dataset	Peak ratio range	Change ratio range	Main observation
PeMS04	1.75–1.99	2.21–2.54	Peaks and changes amplify errors
PeMS08	1.54–2.01	1.72–2.23	Changes are consistently difficult
PeMS07(M)	0.54–0.89	1.85–2.92	Speed peaks are not critical

The critical-context analysis highlights that the notion of difficulty depends on the predicted variable. For PeMS04 and PeMS08, where the target is traffic flow, both peak traffic and large temporal changes increase the error relative to the global MAE. In contrast, for PeMS07(M), where the target is traffic speed, high values are not necessarily difficult because they may correspond to free-flow conditions. Large temporal changes, however, consistently amplify errors across all datasets. This result supports the main argument of the paper: aggregate metrics quantify average error magnitude, but they do not reveal whether errors persist over time or become concentrated in dynamically difficult traffic conditions.

4.3 Spatial and Spectral Residual Organization

We next examine whether residuals are uniformly distributed over the traffic graph or concentrated around specific sensors. Spatial concentration is measured as the ratio between the mean MAE of the top 10% most erroneous nodes and the global MAE. Values greater than one indicate that a small subset of sensors accumulates larger errors than suggested by the aggregate score. Table 4 summarizes this concentration effect together with mean high-frequency residual energy (HFER), used as a global graph-spectral summary of residual irregularity.

Table 4: Summary of spatial and spectral residual organization. Spatial concentration is measured as the ratio between the mean MAE of the top 10% most erroneous nodes and the global MAE. HFER denotes mean high-frequency residual energy.

Dataset	Spatial ratio range	Max spatial ratio	HFER range	Main observation
PeMS04	1.98–2.36	2.36	0.230–0.241	Strong node-level concentration
PeMS07(M)	1.67–1.88	1.88	0.236–0.254	Moderate spatial concentration
PeMS08	1.72–2.92	2.92	0.258–0.271	Strongest concentration

The results show that residual errors are not uniformly distributed over the traffic network. Across all datasets, the top 10% most erroneous nodes exhibit substantially larger errors than the global MAE. This effect is strongest on PeMS08, where the concentration ratio reaches nearly three, but it is also clearly

visible on PeMS04 and PeMS07(M). Therefore, a satisfactory global MAE can still conceal localized failure regions affecting a limited subset of sensors.

The spectral results provide a complementary view. Mean HFER varies less across models than the node-level concentration ratios, indicating that spectral residual energy is less discriminative as a primary ranking criterion in these experiments. While mean HFER provides a global spectral summary, the upper-tail spectral behaviour reveals a useful model-level contrast. In particular, the model with the best global MAE is not always the one with the lowest q_{90} HFER, as summarized in Table 5. This indicates that pointwise accuracy and high-frequency residual irregularity capture different aspects of forecasting behaviour.

Table 5: Model-level contrast between global accuracy and upper-tail spectral irregularity. Lower MAE indicates better pointwise accuracy, whereas lower q_{90} HFER indicates lower high-frequency residual irregularity in the upper tail.

Dataset	Best MAE model	Lowest q_{90} HFER model	Main observation
PeMS04	GWNET	GRU	Best accuracy does not imply lowest spectral tail
PeMS07(M)	AGCRN	ASTGCN	Accurate models keep higher spectral tails
PeMS08	AGCRN	GRU	Spectral tail differs from accuracy ranking

These results show that the spatial structure hidden by aggregate metrics appears in two complementary ways. First, node-level analysis reveals where errors concentrate in the network. Second, spectral analysis indicates whether residuals contain upper-tail high-frequency irregularities that are not captured by MAE or RMSE. The spectral diagnostic is therefore not used as a standalone accuracy criterion; rather, it helps distinguish pointwise accuracy from graph-level residual organization. For the GRU baseline, this interpretation is applied a posteriori: GRU does not explicitly model graph topology, but its prediction errors can still be projected onto the sensor graph to assess how smooth or irregular they are over the network.

4.4 Spatio-Temporal Dynamical Consistency

The previous analyses show that residuals may persist over time and concentrate around specific nodes. We now ask a more direct question: do models that achieve low aggregate errors also preserve the spatio-temporal evolution of the traffic field?

To address this, we use the Graph-Temporal Operator Error (GTOE), introduced in Section 2.4, combining a temporal consistency term and a graph-spatial consistency term based on the Laplacian. Lower GTOE indicates better preservation of traffic dynamics.

Figure 2 reports GTOE across forecasting horizons. Dynamical consistency does not degrade uniformly. On PeMS04, DCRNN and STGCN show a clear horizon-dependent increase, whereas AGCRN and GWNET remain lower and more stable. On PeMS07(M), a similar separation appears. On PeMS08, several

models exhibit an oscillatory pattern across horizons, alternating between over-smoothing and under-smoothing relative to the observed Laplacian signal — a behavior that remains invisible in aggregate MAE or RMSE, which stay stable across the same horizons.

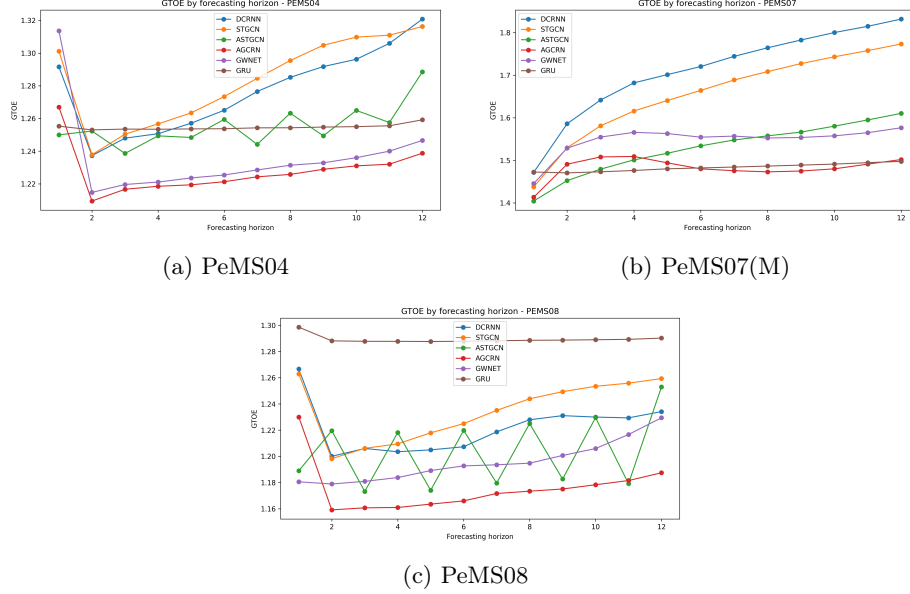


Fig. 2: Graph-Temporal Operator Error across forecasting horizons on PeMS04, PeMS07(M), and PeMS08. Lower values indicate better preservation of temporal changes and graph-spatial variations.

Table 6 reveals two distinct and complementary forms of evidence that aggregate metrics provide an incomplete picture of model behaviour.

The first is a divergence between accuracy and dynamics. On PeMS04, GWNEN obtains the best MAE and RMSE and would be selected under standard evaluation practice. Yet AGCRN achieves substantially lower GTOE, meaning it better preserves the temporal and graph-spatial structure of traffic. A practitioner relying solely on MAE or RMSE would select a model that is less favorable under the proposed dynamical diagnostic, without any indication from the aggregate metrics that this trade-off exists.

The second is an internal inconsistency within aggregate metrics themselves. On PeMS07(M), MAE and RMSE disagree: AGCRN achieves the best MAE while GWNEN achieves the best RMSE, producing conflicting recommendations from two metrics routinely treated as interchangeable. This inconsistency is itself evidence of their limitations. GTOE aligns with MAE on this dataset, suggesting that dynamical consistency captures a dimension closer to average behaviour than to error variance.

On PeMS08, MAE, RMSE, and GTOE all select AGCRN. This agreement shows that the divergences observed on the other datasets are not artifacts

— they are dataset-dependent, which is precisely what a diagnostic framework should reveal. This pattern may be related to the predicted variable. One possible explanation is that traffic speed may exhibit smoother temporal transitions than traffic flow in some settings, although this hypothesis would require further investigation. In contrast, traffic flow may involve sharper transitions and congestion propagation effects, making accuracy and dynamical consistency more likely to diverge.

Table 6: Comparison between aggregate accuracy and dynamical consistency rankings.

Dataset	Best MAE	Best RMSE	Best GTOE	Main observation
PeMS04	GWNET	GWNET	AGCRN	Accuracy and dynamics diverge
PeMS07(M)	AGCRN	GWNET	AGCRN	MAE and RMSE internally disagree
PeMS08	AGCRN	AGCRN	AGCRN	All views agree

Taken together, these results show that aggregate accuracy and dynamical consistency provide complementary views of model behaviour, with GTOE revealing differences that aggregate scores may conceal.

4.5 Ranking Agreement Across Evaluation Views

Finally, we examine whether model rankings remain stable across evaluation views. Table 7 summarizes the best-performing models according to aggregate accuracy, temporal residual dependence, spatial error concentration, and GTOE. For the residual-based diagnostics, lower values indicate more favorable behaviour.

Table 7: Ranking summary across evaluation views.

Dataset	Best MAE	Best RMSE	Best temporal	Best spatial	Best GTOE
PeMS04	GWNET	GWNET	STGCN	STGCN	AGCRN
PeMS08	AGCRN	AGCRN	STGCN	STGCN	AGCRN
PeMS07(M)	AGCRN	GWNET	STGCN	DCRNN	AGCRN

The results show that the best model is not invariant across evaluation criteria. On PeMS04, GWNET is selected by MAE and RMSE, STGCN by temporal and spatial diagnostics, and AGCRN by GTOE. On PeMS08, AGCRN remains best under MAE, RMSE, and GTOE, but not under the temporal and spatial diagnostics. On PeMS07(M), the disagreement is stronger, with different criteria selecting AGCRN, GWNET, STGCN, and DCRNN. This confirms that model selection depends on the aspect of forecasting behaviour being evaluated.

To quantify the agreement between aggregate accuracy and dynamical consistency, Table 8 reports Spearman and Kendall rank correlations between MAE/RMSE rankings and GTOE rankings.

Table 8: Rank correlation between aggregate metrics and GTOE.

Dataset	Spearman MAE	Kendall MAE	Spearman RMSE	Kendall RMSE
PeMS04	0.54	0.33	0.54	0.33
PeMS08	0.71	0.60	0.71	0.60
PeMS07(M)	0.09	0.07	-0.09	-0.07

The correlation results show that agreement between aggregate accuracy and GTOE is dataset-dependent. PeMS08 exhibits the strongest agreement, PeMS04 shows only moderate agreement, and PeMS07(M) shows almost no agreement between MAE and GTOE, with a slightly negative correlation between RMSE and GTOE. These differences are interpreted diagnostically rather than as statistically significant claims, consistently with the single-configuration experimental protocol of this study.

Overall, these results support the main hypothesis of the paper: conventional metrics remain necessary, but they do not fully describe how forecasting errors are structured over time, across sensors, and along the graph. Model selection based only on MAE or RMSE may therefore overlook residual structures that are relevant for assessing forecasting reliability.

5 Conclusion

Accurate model evaluation is a critical prerequisite for progress in spatio-temporal forecasting. Yet, the predominant reliance on aggregate metrics such as MAE and RMSE raises questions about whether these scores fully characterize model behaviour across all relevant dimensions. In this paper, we revisited the evaluation of graph-based traffic forecasting models from a residual-structure perspective. Rather than proposing a new forecasting architecture, we examined whether conventional aggregate metrics such as MAE and RMSE are sufficient to characterize model behavior. Using benchmark PeMS datasets and representative forecasting models, we analyzed prediction residuals across temporal, spatial, spectral, and operator-based dynamical dimensions.

The results show that aggregate metrics provide a necessary but incomplete view of forecasting performance. While MAE and RMSE remain useful for summarizing pointwise accuracy, they do not fully describe how errors persist over time, concentrate across sensors, or affect the preservation of graph-temporal traffic dynamics. In particular, the ranking analysis showed that model selection can change across evaluation views: models selected by aggregate accuracy are not always those favored by temporal, spatial, or dynamical diagnostics. The GTOE analysis further showed that forecasting accuracy and dynamical consistency may agree on some datasets, but diverge on others.

These findings demonstrate the benefit of complementary evaluation: the proposed diagnostics reveal residual structures and ranking trade-offs hidden by aggregate scores, while GTOE distinguishes models by their ability to preserve traffic dynamics. All experiments follow the default LibCity configuration,

including a fixed random seed (seed = 42), ensuring full reproducibility and consistency with the standard evaluation protocol of the framework. Future work will extend this framework toward context-aware and uncertainty-aware evaluation, multi-seed variance estimation, and the integration of residual-structure diagnostics into model selection and training objectives.

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